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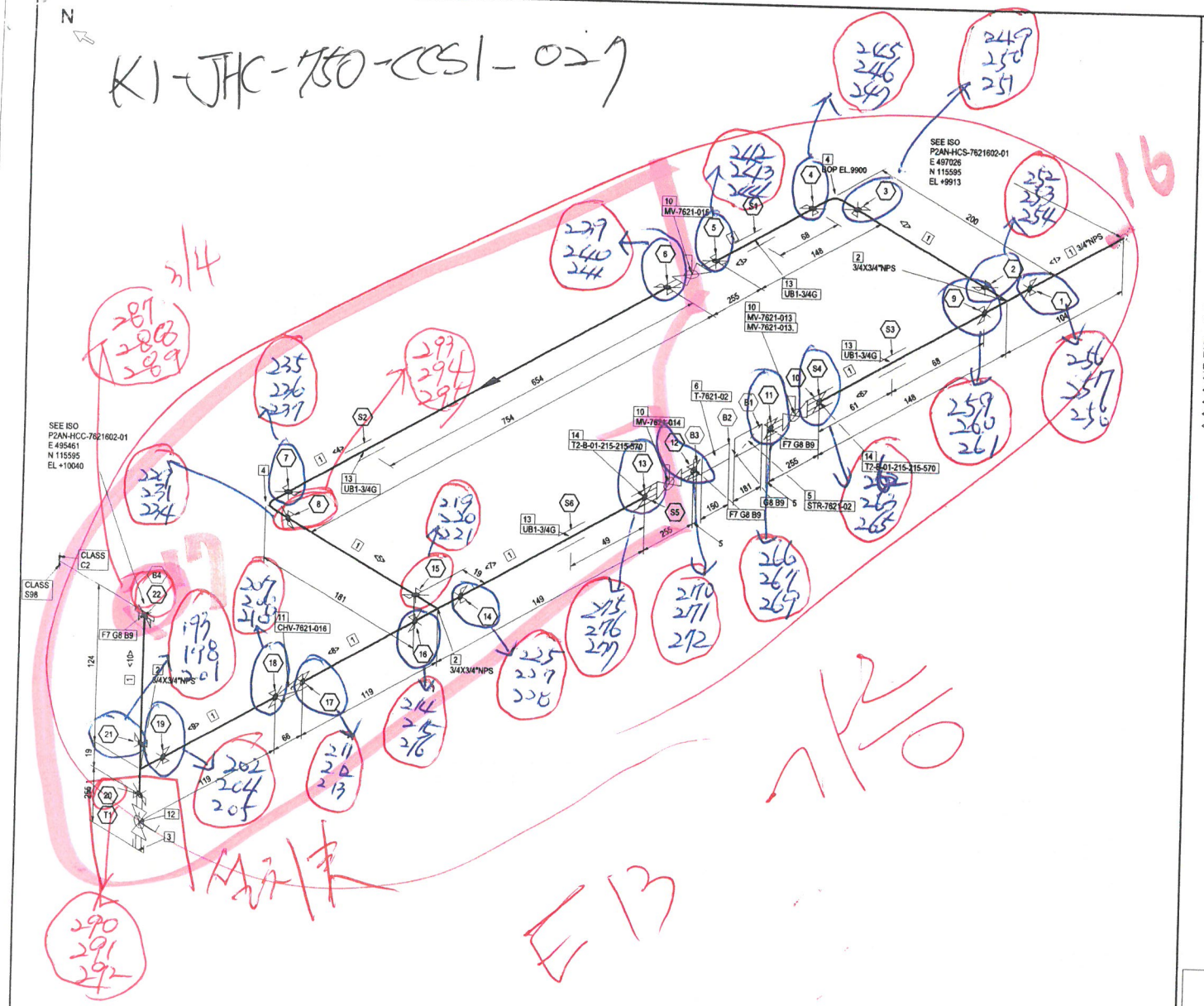

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (°c)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (°c)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
0.75	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
1	19-09-25	ISSUED FOR REVISION	J.Y.KIM	M.H.SUNG	T.H.LIM
0	29-08-25	ISSUED FOR INFORMATION	J.Y.KIM	M.H.SUNG	T.H.LIM

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BILL OF MATERIALS

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE SMLS ASTM A312-316/316L SCH40S PE	3/4"	1.9 M	S98	P	PA02-000000956
2	TEE SW CL3000 ASTM A182-F316/316L	3/4X3/4"	3	S98	T	PB99-000038722
3	CAP SCRD (NPT) CL3000 ASTM A182-F316/316L	3/4"	1	S98	CPT	PB99-000038501
4	90 DEG. ELBOW (LR) SW CL3000 ASTM A182-F316/316L	3/4"	2	S98	E	PB99-000038441
5	TAG NO : STR-7621-02, STRAINER - Y-TYPE	3/4"	1			SPECIALTY
6	TAG NO : T-7621-02, STEAM TRAP - LINE TRAP	3/4"	1			SPECIALTY
7	SOCKET-WELDING FLANGE RF CL150 ASTM A182-F316/316L SCH40S	3/4"	3	S98	F	PB01-000004840
8	SPIRAL WOUND GASKET O-RING: SS 304 CL150 HOOP : SS 316L FILLER : GRAPHIT E 4.5MM(0.18") THK.	3/4"	4	S98	G	PF01-000009264
9	70 STUD BOLT W/2-NUT HEAVY HEX. NUT ASTM A193-B8 CL.2 ASTM A194-8 UNI/UN R x MM	1/2"	16	S98	B	PB99-000076882
10	BALL VALVE SW ANSI(ASTM) CL600 ASTM A351-CF8M SS316 TRIM RTFE SEAT FLO ATING BALL FB 3 PIECE BODY LV. OP. (VAB060T)	3/4"	3	S98	VL	PP04-000007760
11	CHECK VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BC PISTON DISC (VAC015)	3/4"	1	S98	VC	PP06-000009507
12	BALL VALVE SW W/NIPPLE ANSI(ASTM) CL600 ASTM A351-CF8M SS316 TRIM RTFE SEAT FLOATING BALL FB 3 PIECE BODY LV. OP. A312-316/316L SCH80S 75MM TOE (NPT)/POE (VL1-D16)	3/4"	1	S98	VLST1	PP04-000006192
13	UB1-3/4G	3/4"	4			UB1-G
14	T2-B-01-215-215-570	3/4"	2			T2-B

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	65	3/4		<2>	162	3/4	
<3>	129	3/4		<4>	735	3/4	
<5>	162	3/4		<6>	129	3/4	
<7>	130	3/4		<8>	100	3/4	
<9>	100	3/4		<10>	100	3/4	

F3047

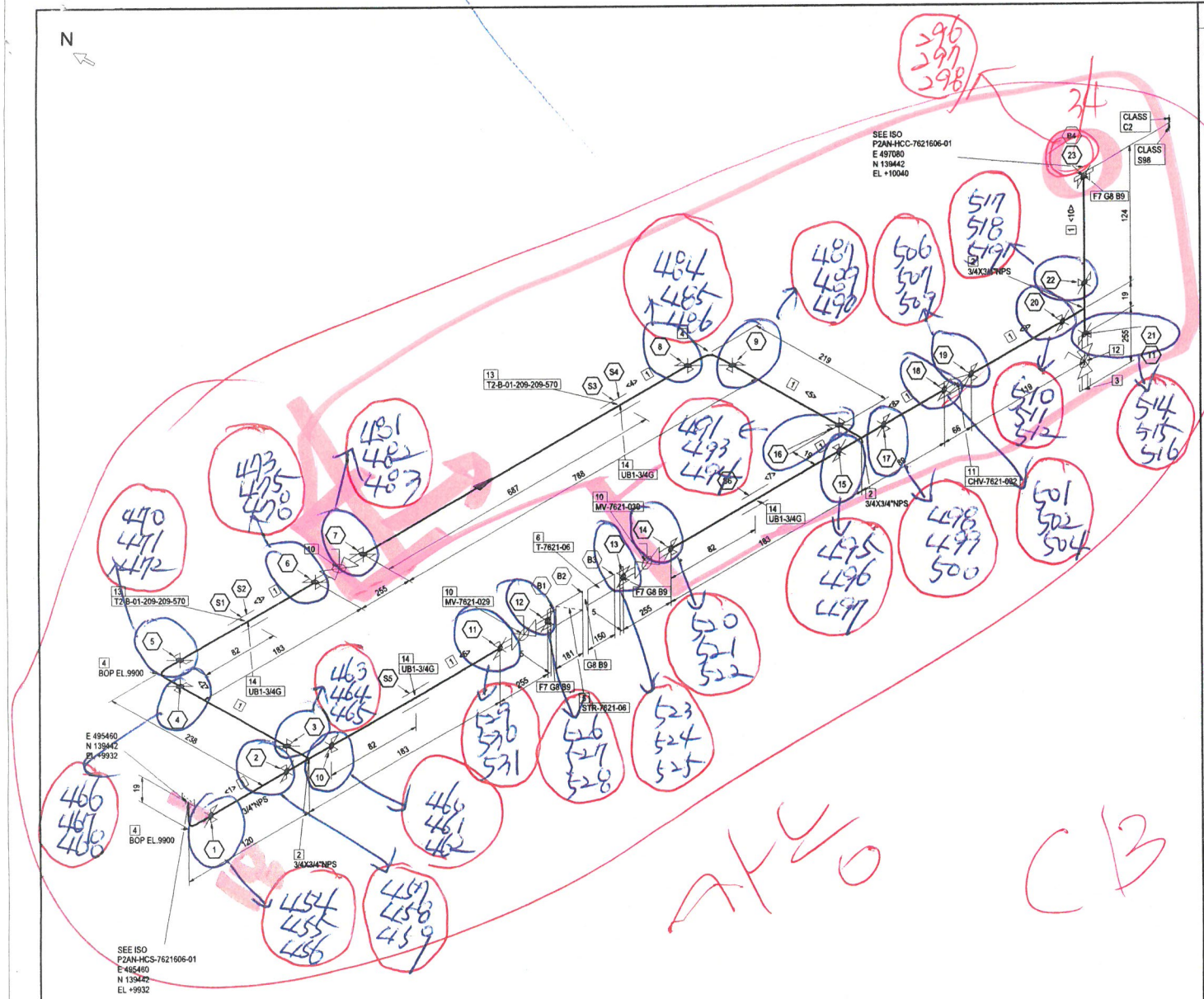
Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-09-19

PROJECT
PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB
P&ID NO.	K1-DSL-UT-PID-7621-01			CWA NO.	P2AN
GAD NO.	-			PROJ. NO.	16480
				FLUID	HCS
				LINE NO.	0.75-HCS-7621602-S98-IHI
DWG NO.	P2AN-HCS-7621602-02			SH'T NO.	1 OF 1
				REV	1

PIPE modified at 15:59:27 Aug 2025, Drawing Created at 18:15:07 27 August 2025
PIPE NAME : 0.75-HCS-7621602-S98-IHI_P2AN-02



BILL OF MATERIALS

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE SMLS ASTM A312-316/316L SCH40S PE	3/4"	2.1 M	S98	P	PA02-000000956
2	TEE SW CL3000 ASTM A182-F316/316L	3/4X3/4"	3	S98	T	PB99-000038722
3	CAP SCRD (NPT) CL3000 ASTM A182-F316/316L	3/4"	1	S98	CPT	PB99-000038501
4	90 DEG. ELBOW (LR) SW CL3000 ASTM A182-F316/316L	3/4"	3	S98	E	PB99-000038441
5	TAG NO : STR-7621-06, STRAINER - Y-TYPE	3/4"	1			SPECIALTY
6	TAG NO : T-7621-06, STEAM TRAP - LINE TRAP	3/4"	1			SPECIALTY
7	SOCKET-WELDING FLANGE RF CL150 ASTM A182-F316/316L SCH40S	3/4"	3	S98	F	PB01-000004840
8	SPIRAL WOUND GASKET O-RING: SS 304 CL150 HOOP : SS 316L FILLER : GRAPHIT E 4.5MM(0.18") THK.	3/4"	4	S98	G	PF01-000009284
9	70 STUD BOLT W/2-NUT HEAVY HEX, NUT ASTM A193-B8 CL.2 ASTM A194-8 UNUN R'x MM	1/2"	16	S98	B	PB99-000076882
10	BALL VALVE SW ANSI(ASTM) CL600 ASTM A351-CF8M SS316 TRIM RTFE SEAT FLO ATING BALL FB 3 PIECE BODY LV. OP. (VAB060T)	3/4"	3	S98	VL	PP04-000007760
11	CHECK VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BC PISTON DISC (VAC015)	3/4"	1	S98	VC	PP06-000009507
12	BALL VALVE SW W/NIPPLE ANSI(ASTM) CL600 ASTM A351-CF8M SS316 TRIM RTFE SEAT FLOATING BALL FB 3 PIECE BODY LV. OP. A312-316/316L SCH80S 75MM TOE (NPT)/POE (VL1-D16)	3/4"	1	S98	VLST1	PP04-000008192
13	T2-B-01-209-209-570	3/4"	2			T2-B
14	UB1-3/4G	3/4"	4			UB1-G

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	82	3/4		<2>	200	3/4	
<3>	163	3/4		<4>	768	3/4	
<5>	200	3/4		<6>	163	3/4	
<7>	163	3/4		<8>	70	3/4	
<9>	100	3/4		<10>	100	3/4	

F2712

Project PROVIDENCE

Approval For Construction
PIPING MANAGER Bea Chang Won
2025-09-13

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRFSS. (bar.g)	OPER. TEMP. (°c)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (°c)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
0.75	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
0	13-09-25	ISSUED FOR INFORMATION	J.Y.KIM	M.H.SUNG	T.H.LIM

PROJECT: PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB	CWA NO.	P2AN
P&ID NO.	K1-DSL-UT-PID-7621-01			PROJ. NO.	16480	FLUID	HCS
CAD NO.	-			LINE NO.	0.75-HCS-7621606-S98-IHI		
DWG NO.	P2AN-HCS-7621606-02					SH'T NO. 1 OF 1	REV 0

PIPE modified at 14:21 28 Aug 2025 , Drawing Created at 14:39:20 28 August 2025
PIPE NAME : 0.75-HCS-7621606-S98-IHI_P2AN-02

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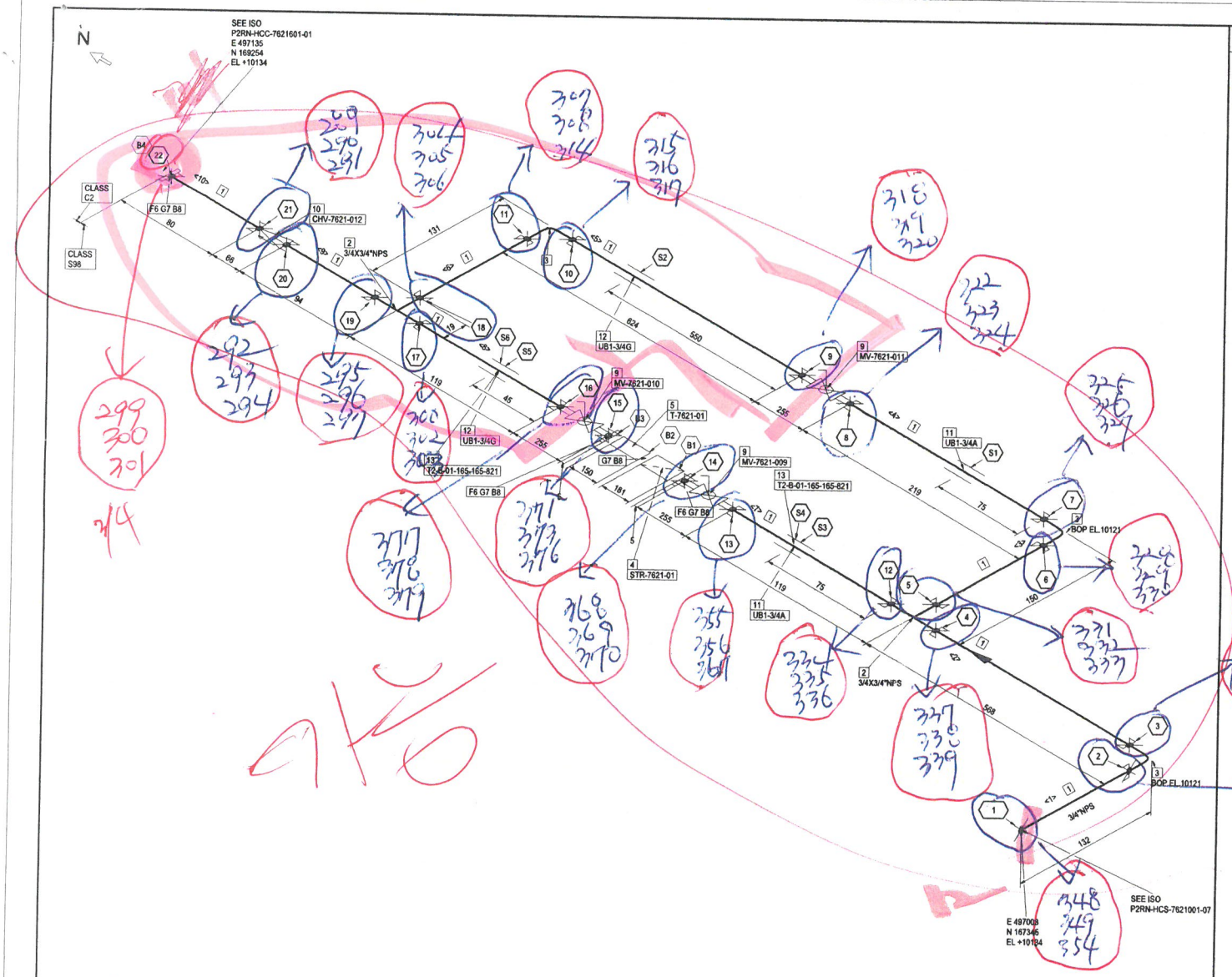

BILL OF MATERIALS

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE SMLS ASTM A312-316/316L SCH40S PE	3/4"	2.1 M	S98	P	PA02-000000956
2	TEE SW CL3000 ASTM A182-F316/316L	3/4X3/4"	2	S98	T	PB99-000038722
3	90 DEG. ELBOW (LR) SW CL3000 ASTM A182-F316/316L	3/4"	3	S98	E	PB99-000038441
4	TAG NO : STR-7621-01, STRAINER - Y-TYPE	3/4"	1			SPECIALTY
5	TAG NO : T-7621-01, STEAM TRAP - LINE TRAP	3/4"	1			SPECIALTY
6	SOCKET-WELDING FLANGE RF CL150 ASTM A182-F316/316L SCH40S	3/4"	3	S98	F	PB01-000004840
7	SPIRAL WOUND GASKET O-RING: SS 304 CL150 HOOP : SS 316L FILLER : GRAPHIT E 4.5MM(0.18") THK.	3/4"	4	S98	G	PF01-000009264
8	70 STUD BOLT W/2-NUT HEAVY HEX. NUT ASTM A193-B8 CL.2 ASTM A194-8 UNUN R x MM	1/2"	16	S98	B	PB99-000076882
9	BALL VALVE SW ANSI(ASTM) CL600 ASTM A351-CF8M SS316 TRIM RTFE SEAT FLO ATING BALL FB 3 PIECE BODY LV, OP, (VAB060T)	3/4"	3	S98	VL	PP04-000007760
10	CHECK VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BC PISTON DISC (VAC015)	3/4"	1	S98	VC	PP06-000009507
11	UB1-3/4A	3/4"	2			UB1-A
12	UB1-3/4G	3/4"	2			UB1-G
13	T2-B-01-165-165-821	3/4"	2			T2-B

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	113	3/4		<2>	530	3/4	
<3>	112	3/4		<4>	200	3/4	
<5>	805	3/4		<6>	112	3/4	
<7>	100	3/4		<8>	100	3/4	
<9>	75	3/4		<10>	75	3/4	

F2351

Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-08-29

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (°C)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (°C)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
0.75	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-

REV	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
0	29-08-25	ISSUED FOR INFORMATION	J.Y.KIM	M.H.SUNG	T.H.LIM

PROJECT: PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB
P&ID NO.	K1-DSL-UT-PID-7621-01			PROJ. NO.	16480
CAD NO.	-			LINE NO.	0.75-HCS-7621601-S98-IHI
DWG NO.	P2RN-HCS-7621601-01			SH'T NO.	1 OF 1

CWA NO. P2RN
FLUID HCS
REV. 0

PIPE modified at 16:02:28 Jul 2025, Drawing Created at 19:13:52 31 July 2025
PIPE NAME: 0.75-HCS-7621601-S98-IHI_P2RN-01

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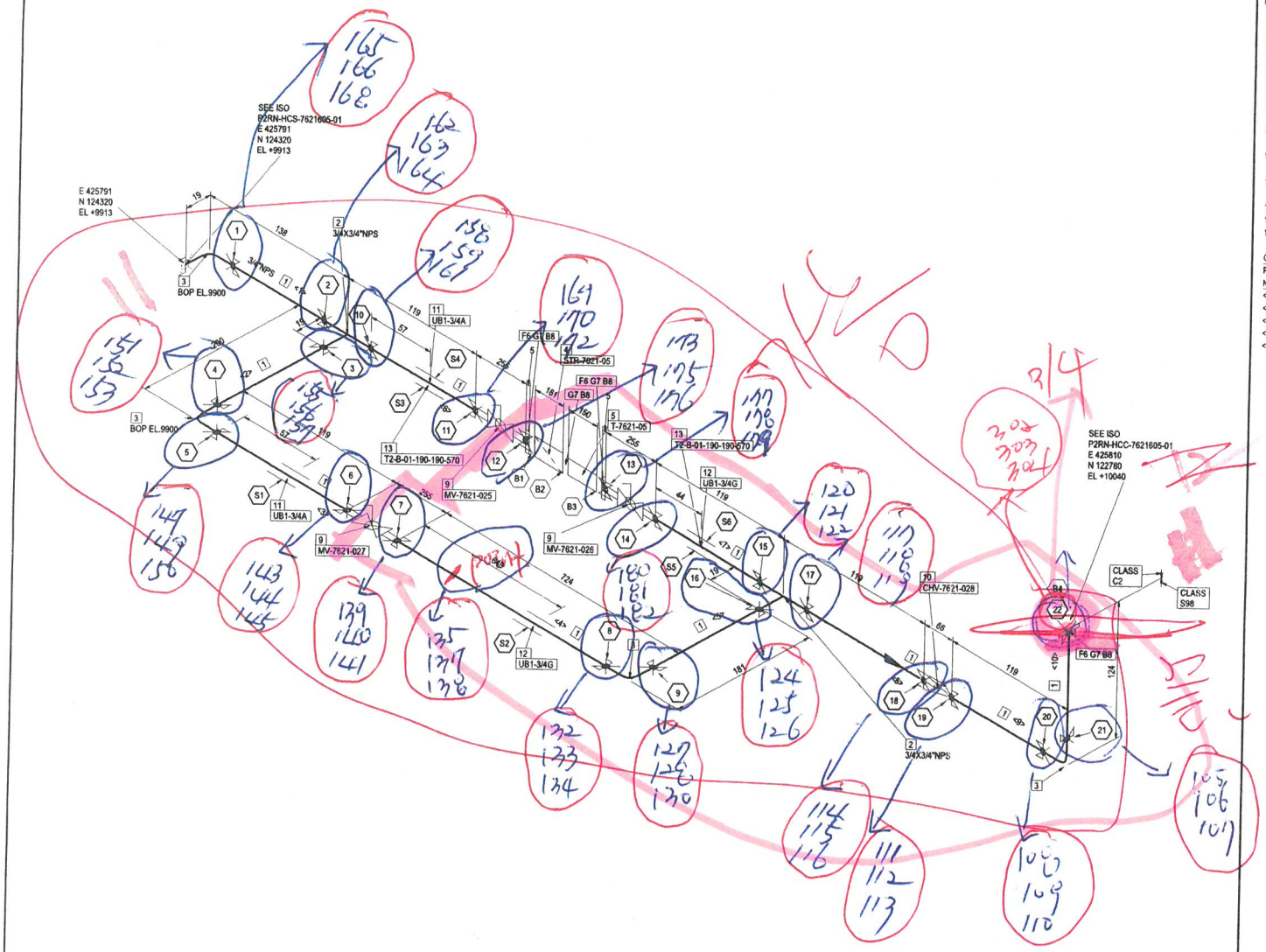

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (°c)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (°c)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
0.75	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
0	29-08-25	ISSUED FOR INFORMATION	J.Y.KIM	M.H.SUNG	T.H.LIM

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BILL OF MATERIALS

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE SMLS ASTM A312-316/316L SCH40S PE	3/4"	1.8 M	S98	P	PA02-000000956
2	TEE SW CL3000 ASTM A182-F316/316L	3/4X3/4"	2	S98	T	PB99-000038722
3	90 DEG. ELBOW (LR) SW CL3000 ASTM A182-F316/316L	3/4"	4	S98	E	PB99-000038441
4	TAG NO : STR-7621-05, STRAINER - Y-TYPE	3/4"	1			SPECIALTY
5	TAG NO : T-7621-05, STEAM TRAP - LINE TRAP	3/4"	1			SPECIALTY
6	SOCKET-WELDING FLANGE RF CL150 ASTM A182-F316/316L SCH40S	3/4"	3	S98	F	PB01-000004840
7	SPIRAL WOUND GASKET O-RING: SS 304 CL150 HOOP : SS 316L FILLER : GRAPHIT E 4.5MM(0.18") THK.	3/4"	4	S98	G	PF01-000009264
8	70 STUD BOLT W/2-NUT HEAVY HEX. NUT ASTM A193-B8 CL.2 ASTM A194-8 UNUN R x MM	1/2"	16	S98	B	PB99-000076882
9	BALL VALVE SW ANSI(ASTM) CL600 ASTM A351-CF8M SS316 TRIM RTFE SEAT FLO ATING BALL FB 3 PIECE BODY LV. OP. (VAB060T)	3/4"	3	S98	VL	PP04-000007760
10	CHECK VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BC PISTON DISC (VAC015)	3/4"	1	S98	VC	PP06-000009507
11	UB1-3/4A	3/4"	2			UB1-A
12	UB1-3/4G	3/4"	2			UB1-G
13	T2-B-01-190-190-570	3/4"	2			T2-B

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	100	3/4		<2>	162	3/4	
<3>	100	3/4		<4>	705	3/4	
<5>	162	3/4		<6>	100	3/4	
<7>	100	3/4		<8>	100	3/4	
<9>	100	3/4		<10>	100	3/4	

F2353

Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-08-29

PROJECT
PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB
P&ID NO.	K1-DSL-UT-PID-7621-01			PROJ. NO.	16480
CAD NO.	-			LINE NO.	0.75-HCS-7621605-S98-IHI
DWG NO.	P2RN-HCS-7621605-02			SH'T NO.	1 OF 1

CWA NO. P2RN
FLUID HCS
REV. 0

PIPE modified at 10:33:29 Jul 2025 , Drawing Created at 19:13:58 31 July 2025
PIPE NAME : 0.75-HCS-7621605-S98-IHI_P2RN-02